

_03-regression-graph

RStudio assignment · Team SAUGE / LAGARDE / AHMAD

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8 October 2026

Think before coding (1.1)

Scientific question. Is the effect of task difficulty (ID) on movement time the same in the three groups?

1. **How do we visualise it?** Mean MT ($\pm$ SD) as a function of ID, one line per group. If the effect of ID is the same in all groups, the lines are parallel.
2. **Is there a significant difference?** In a mixed ANOVA, the effect of ID differs between groups if the **GROUP $\times$ ID interaction** is significant. The main effects of GROUP and ID are secondary, and must be read with care if the interaction is significant [1].
3. **What sentence do we write?** A sentence that gives the direction of the effect and the statistics, F(DFn, DFd), p and an effect size.

Which library? (1.2)

We use **ez** [2], the library suggested first in the assignment. Its three functions cover the whole analysis with the same arguments (data, dependent variable, participant identifier, between- and within-subjects factors):

- **ezANOVA()** runs the mixed ANOVA and returns the ANOVA table with the generalised eta squared (*ges*) as effect size [3], [4] and, when needed, Mauchly's test of sphericity with the Greenhouse–Geisser and Huynh–Feldt corrections;
- **ezPlot()** plots the cell means, with error bars that show which means differ;
- **ezStats()** returns the descriptive statistics (N, mean, SD) of each cell.

Data reorganisation and figures use the **tidyverse** [5].

Data reorganisation and ANOVA (1.3)

Load the data

Each row of **Results.txt** is one movement: group, participant (**SUBJ**), trial, orientation (**ORI**), direction (**DIR**), index of difficulty (**ID**), repetition (**REP**), number of velocity peaks (**NbVelPeaks**, i.e. the number of sub-movements) and movement time in seconds.

```
library(tidyverse) # data wrangling and ggplot2
library(ez)        # mixed-design ANOVA (ezANOVA, ezPlot, ezStats)
library(here)      # paths relative to the project root
theme_set(theme_minimal(base_size = 11))
```

```
raw <- read.table(here("data", "Results.txt"), header = TRUE,
                  stringsAsFactors = TRUE)

glimpse(raw)
```

```
Rows: 7,560
Columns: 9
$ GROUP      <fct> ap, ap, ap, ap, ap, ap, ap, ap, ap, ap, ap, ap, ap, ap, ap, a...
$ SUBJ       <fct> apS01, apS01, apS01, apS01, apS01, apS01, apS01, apS01, a...
$ TRIAL      <fct> T01, T01, T01, T01, T01, T01, T01, T01, T01, T01, T01, T0...
$ ORI        <fct> W, W, NW, NW, N, N, NE, NE, E, E, W, W, NW, NW, N, N, NE,...
$ DIR        <fct> outward, inward, outward, inward, outward, inward, outwar...
$ ID         <fct> ID1, ID1, ID1, ID1, ID1, ID1, ID1, ID1, ID1, ID1, ID1, ID2, ID...
$ REP        <fct> R1, R1, R1, R1, R1, R1, R1, R1, R1, R1, R1, R1, R1, R1, R1, R...
$ NbVelPeaks <dbl> 2, 2, 3, 2, 3, 3, 1, 2, 4, 1, 2, 6, 3, 2, 4, 5, 6, 3, 3, ...
$ MovementTime <dbl> 1.23, 1.26, 1.14, 0.90, 1.11, 0.81, 1.08, 0.81, 1.32, 1.0...
```

Check for missing data

A mixed ANOVA needs every participant to have a value in every cell of the within-subjects factors. We count the movements in each combination of factors.

```
raw |> count(GROUP, ID)
```

	GROUP	ID	n
1	ap	ID1	740
2	ap	ID2	740
3	ap	ID3	740
4	cp	ID1	1140
5	cp	ID2	1140
6	cp	ID3	1140
7	pp	ID1	760
8	pp	ID2	760
9	pp	ID3	400

The group pp has only 400 movements for ID3, against 760 for ID1 and ID2: many patients did not perform ID3, the most difficult condition. We remove ID3.

```
d <- raw |> filter(ID != "ID3") |> droplevels()
levels(d$ID)
```

```
[1] "ID1" "ID2"
```

```
d |> count(GROUP, ID)
```

	GROUP	ID	n
1	ap	ID1	740
2	ap	ID2	740
3	cp	ID1	1140
4	cp	ID2	1140
5	pp	ID1	760
6	pp	ID2	760

We now check the repetitions.

```
d |> count(REP)
```

	REP	n
1	R1	2040
2	R2	2040
3	R3	1200

```
d |> count(GROUP, REP)
```

	GROUP	REP	n
1	ap	R1	520
2	ap	R2	520
3	ap	R3	440
4	cp	R1	760
5	cp	R2	760
6	cp	R3	760
7	pp	R1	760
8	pp	R2	760

The third repetition is missing for the group `pp` and incomplete for the group `ap` (440 instead of 520 movements). We remove R3.

```
d <- d |> filter(REP != "R3") |> droplevels()
d |> count(GROUP, REP)
```

	GROUP	REP	n
1	ap	R1	520
2	ap	R2	520
3	cp	R1	760
4	cp	R2	760
5	pp	R1	760
6	pp	R2	760

```
d |> distinct(GROUP, SUBJ) |> count(GROUP, name = "count")
```

	GROUP	count
1	ap	13
2	cp	19
3	pp	19

The final data set has 13 age-matched adults, 19 young adults and 19 patients (51 participants), which matches the 48 error degrees of freedom of the article ($51 - 3 \text{ groups} = 48$).

One value per participant and condition

The ANOVA must compare participants, not movements: otherwise the hundreds of movements of one participant would be treated as independent observations (pseudo-replication). We average the movements of each participant in each ID.

```
mt <- d |>
  group_by(GROUP, SUBJ, ID) |>
  summarise(MT = mean(MovementTime), .groups = "drop")
mt |> count(GROUP, ID)
```

```
# A tibble: 6 × 3
  GROUP ID      n
  <fct> <fct> <int>
1 ap    ID1     13
2 ap    ID2     13
3 cp    ID1     19
4 cp    ID2     19
5 pp    ID1     19
6 pp    ID2     19
```

Run the ANOVA

`ezANOVA()` takes the dependent variable (`dv`), the participant identifier (`wid`), the between-subjects factor and the within-subjects factor. We ask for type III sums of squares because the groups have unequal sizes (13, 19 and 19 participants).

```
aov_mt <- ezANOVA(data = mt, dv = MT, wid = SUBJ,
                  between = GROUP, within = ID, type = 3)
aov_mt$ANOVA
```

	Effect	DFn	DFd		F	p	p<.05	ges
2	GROUP	2	48	37.24408	1.720007e-10	*	0.57462110	
3	ID	1	48	26.10886	5.543011e-06	*	0.06581307	
4	GROUP:ID	2	48	11.58124	7.866206e-05	*	0.05882293	

```
aov_tab <- aov_mt$ANOVA
```

With only two levels, the within-subjects factor ID cannot violate sphericity, so `ez` does not report Mauchly's test and no Greenhouse–Geisser correction is needed.

Visualise the interaction

`ezPlot()` draws the cell means. Its error bars are Fisher's Least Significant Difference (FLSD) bars: two means whose bars do not overlap differ significantly.

```
ezPlot(data = mt, dv = MT, wid = SUBJ, between = GROUP, within = ID,
       x = ID, split = GROUP, type = 3,
       x_lab = "ID", y_lab = "Movement Time (s)")
```

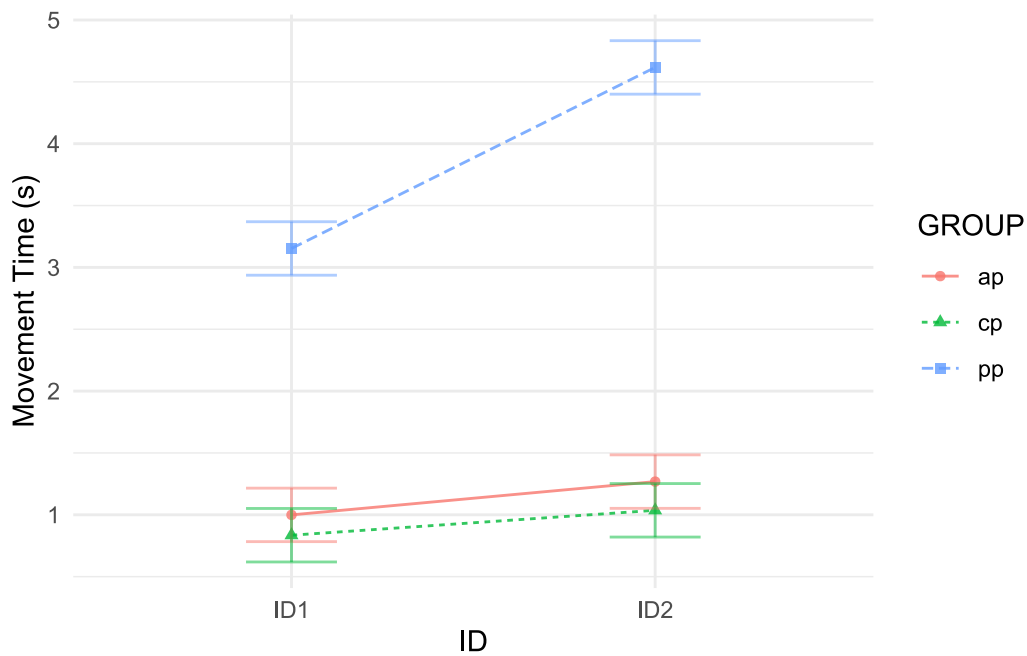


Figure 1: Movement time as a function of ID in each group (ezPlot; error bars: Fisher's Least Significant Difference). The lines are clearly not parallel: the effect of ID is much larger for the patients (pp).

`ezStats()` gives the descriptive statistics of each cell (number of participants, mean, SD and FLSD). We keep them for the next questions.

```
mt_summary <- ezStats(data = mt, dv = MT, wid = SUBJ,
                      between = GROUP, within = ID, type = 3) |>
  rename(n = N, mean = Mean, sd = SD) |>
  as_tibble()
mt_summary |> mutate(across(c(mean, sd, FLSD), \(x) round(x, 2)))
```

```
# A tibble: 6 × 6
  GROUP ID      n mean  sd FLSD
<fct> <fct> <dbl> <dbl> <dbl> <dbl>
1 ap    ID1      13  1    0.2  0.43
2 ap    ID2      13 1.27 0.29 0.43
3 cp    ID1      19 0.83 0.12 0.43
4 cp    ID2      19 1.04 0.11 0.43
5 pp    ID1      19 3.15 1.47 0.43
6 pp    ID2      19 4.62 2.41 0.43
```

Are the assumptions met?

`ezANOVA()` reports Levene's test only for purely between-subjects designs, so we run it at each ID with `car::leveneTest()` (installed with `ez`). We check normality in each cell with the Shapiro-Wilk test.

```
mt |> # equal variances between groups
  group_by(ID) |>
  group_modify(\(df, key) {
    lt <- car::leveneTest(MT ~ GROUP, data = df)
    tibble(F = lt$`F value`[1], DFn = lt$Df[1], DFd = lt$Df[2], p = lt$`Pr(>F)`[1])
  })
```

```
# A tibble: 2 × 5
# Groups:   ID [2]
  ID      F  DFn  DFd      p
<fct> <dbl> <int> <int>   <dbl>
1 ID1    21.8     2    48 0.000000184
2 ID2    12.2     2    48 0.0000524
```

```
mt |> # normality in each cell
  group_by(GROUP, ID) |>
  summarise(W = shapiro.test(MT)$statistic, p = shapiro.test(MT)$p.value,
            .groups = "drop")
```

```
# A tibble: 6 × 4
  GROUP ID      W      p
<fct> <fct> <dbl> <dbl>
1 ap   ID1    0.926 0.306
2 ap   ID2    0.785 0.00457
3 cp   ID1    0.960 0.580
4 cp   ID2    0.922 0.122
5 pp   ID1    0.945 0.321
6 pp   ID2    0.881 0.0223
```

The variances are not equal between groups (Levene's test, $p < .001$ at both IDs; the SD of the patients is more than ten times that of the young adults), and movement time departs from normality in two of the six cells (ID2 for the age-matched adults and for the patients, Shapiro-Wilk $p < .05$). With unequal group sizes, this makes the p-values approximate. However, the effects are so large ($p < .001$ for all three effects) that this does not change the conclusion; a log transformation of MT, which reduces the skewness, would be a reasonable robustness check.

Result sentence

Movement time increased with the index of difficulty ($F(1, 48) = 26.11$, $p < .001$, $\eta_G^2 = 0.07$) and was longer for the patients than for both control groups ($F(2, 48) = 37.24$, $p < .001$, $\eta_G^2 = 0.57$). Crucially, the effect of difficulty was not the same in all groups (GROUP $\times$ ID interaction, $F(2, 48) = 11.58$, $p < .001$, $\eta_G^2 = 0.06$): from ID1 to ID2, movement time increased by 1.47 s in the patients (3.15 to 4.62 s), but only by 0.27 s in the age-matched adults and 0.21 s in the young adults.

Run the linear regression by GROUP

To investigate the relationship between movement time and task difficulty, we fit one linear regression for each group. The dependent variable is movement time (**MT**) and the predictor is the index of difficulty (**ID**).

The regression is performed on the participant-level means created in the previous section. This avoids treating the repeated movements of one participant as independent observations

Prepare the data

ID is a categorical variable with values **ID1** and **ID2**. We create a numeric version for the regression, where **ID1** becomes 1 and **ID2** becomes 2.

```
#| label: regression-data

mt_regression <- mt |>
  mutate(
    ID_numeric = as.numeric(sub("ID", "", ID))
  )

mt_regression
```

```
# A tibble: 102 × 5
  GROUP SUBJ ID      MT ID_numeric
  <fct> <fct> <fct> <dbl>      <dbl>
1 ap    apS01 ID1    1.01         1
2 ap    apS01 ID2    1.30         2
3 ap    apS02 ID1    1.05         1
4 ap    apS02 ID2    1.34         2
5 ap    apS03 ID1    0.835        1
6 ap    apS03 ID2    1.17         2
7 ap    apS04 ID1    1.21         1
8 ap    apS04 ID2    1.58         2
9 ap    apS05 ID1    1.49         1
10 ap   apS05 ID2    2.10         2
# i 92 more rows
```

Fit one regression per group

```
regression_models <- mt_regression |>
  group_by(GROUP) |>
  group_modify(\(df, key) {
    model <- lm(MT ~ ID_numeric, data = df)

    tibble(
      intercept = unname(coef(model)[1]),
      slope = unname(coef(model)[2]),
      r_squared = summary(model)$r.squared,
      p_value = summary(model)$coefficients[2, 4]
    )
  }) |>
  ungroup()
```

regression_models

```
# A tibble: 3 × 5
  GROUP intercept slope r_squared p_value
<fct>    <dbl> <dbl>    <dbl>    <dbl>
1 ap      0.730 0.269     0.235 0.0120
2 cp      0.634 0.201     0.447 0.00000451
3 pp      1.69  1.46      0.124 0.0301
```

The regression equation is:

[$MT = a + b \text{ ID}$]

where (a) is the intercept and (b) is the slope. The slope indicates the increase in movement time associated with a one-unit increase in task difficulty. Therefore, the group with the largest slope is the group most affected by increasing task difficulty.

```
equations <- regression_models |>
  mutate(
    equation = sprintf(
      "MT = %.3f %s %.3f x ID",
      intercept,
      if_else(slope >= 0, "+", "-"),
      abs(slope)
    )
  )
```

```
equations |>
  select(GROUP, equation, r_squared, p_value)
```

```
# A tibble: 3 × 4
  GROUP equation          r_squared    p_value
<fct> <chr>          <dbl>    <dbl>
1 ap    MT = 0.730 + 0.269 x ID  0.235 0.0120
2 cp    MT = 0.634 + 0.201 x ID  0.447 0.00000451
3 pp    MT = 1.689 + 1.464 x ID  0.124 0.0301
```

The regression slopes can be compared with the ANOVA interaction. A significant $\text{GROUP} \times \text{ID}$ interaction means that the effect of difficulty differs between groups. The group with the steepest regression line should show the largest increase in movement time from ID1 to ID2.

Save a graph in a PDF file for publication

The following graph shows the participant-level mean movement times, the regression line for each group, and the equation of each regression.

```
publication_plot <- ggplot(
  mt_regression,
  aes(
    x = ID_numeric,
    y = MT,
    colour = GROUP,
    group = GROUP
  )
) +
  geom_point(
    alpha = 0.25,
    size = 1.8
  ) +
  geom_smooth(
    method = "lm",
    formula = y ~ x,
    se = FALSE,
    linewidth = 0.9
  ) +
  geom_text(
    data = equation_labels,
    aes(
      x = x,
      y = y,
```

```

    label = equation,
    colour = GROUP
  ),
  inherit.aes = FALSE,
  hjust = 0,
  show.legend = FALSE
) +
geom_blank(
  data = tibble(
    ID_numeric = c(0, 3),
    MT = c(0, 6)
  ),
  aes(x = ID_numeric, y = MT),
  inherit.aes = FALSE
) +
scale_x_continuous(
  breaks = c(1, 2),
  labels = c("ID1", "ID2"),
  limits = c(0, 3)
) +
scale_y_continuous(
  limits = c(0, 6)
) +
labs(
  title = "Movement time as a function of task difficulty",
  x = "Index of difficulty (ID)",
  y = "Mean movement time (s)",
  colour = "Group"
) +
theme_classic() +
theme(
  plot.title = element_text(hjust = 0.5),
  legend.position = "right"
)

publication_plot

```

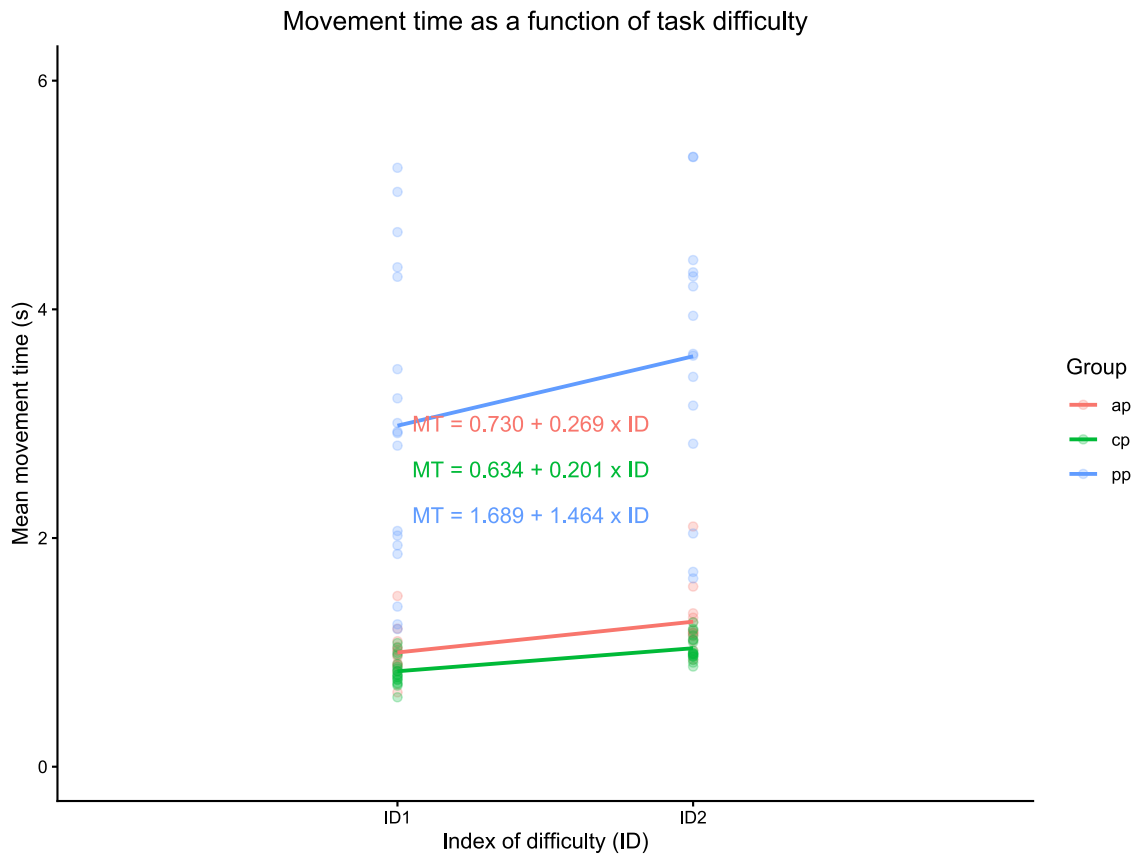


Figure 2: Mean movement time as a function of index of difficulty, with separate linear regression lines and equations for the three groups.

The lines are not parallel: the increase in movement time between ID1 and ID2 is larger for some groups (the stroke survivors) than for others (the healthy age-matched and young participants). This graphical result illustrates the significant $\text{GROUP} \times \text{ID}$ interaction found in the ANOVA.

Export the graph as a PDF file for publication

The graph is exported with a width of 8 inches and a height of 6 inches, as requested in the assignment.

```
ggsave(
  filename = "movement_time_regression.pdf",
  plot = publication_plot,
  width = 8,
  height = 6,
  units = "in",
  device = cairo_pdf
)
```

Differences with the article (1.7)

Putting our equations on the same scale as the article

In the regression section, ID was coded 1 for ID1 and 2 for ID2 (`ID_numeric`). The slopes of `regression_models` are therefore in seconds **per ID step**, and the intercepts are the movement times predicted at a step “0” that does not exist. The article fits Fitts’ law on the index of difficulty in **bits** [6], [7]. The two IDs of the task are 3.80 and 5.43 bits (x-axis of Fig 3 of the article), so one ID step is 1.63 bits.

To compare our equations with those of the article, we convert them to bits. The line goes through the same two group means; only the unit of the x-axis changes:

- slope (s/bit) = slope (s/step) / 1.63;
- intercept (s) = predicted MT at ID1 – slope (s/bit) × 3.80.

```
id_bits <- c(ID1 = 3.80, ID2 = 5.43)
step_bits <- unname(id_bits["ID2"] - id_bits["ID1"])

article <- tibble(
  GROUP = c("pp", "ap", "cp"),
  intercept_article = c(0.422, 0.622, 0.475),
  slope_article     = c(0.672, 0.118, 0.097)
)

comparison <- regression_models |>
  mutate(GROUP = as.character(GROUP),
         slope_ours = slope / step_bits,
         intercept_ours = (intercept + slope * 1) - slope_ours * id_bits["ID1"])
|>
  select(GROUP, intercept_ours, slope_ours, r_squared) |>
  left_join(article, by = "GROUP") |>
  mutate(GROUP = factor(GROUP, levels = c("pp", "ap", "cp"))) |>
  arrange(GROUP) |>
  mutate(slope_ratio_ours = slope_ours / slope_ours[GROUP == "cp"],
         slope_ratio_article = slope_article / slope_article[GROUP == "cp"])

comparison |> mutate(across(where(is.numeric), \(x) round(x, 3)))
```

```
# A tibble: 3 × 8
  GROUP intercept_ours slope_ours r_squared intercept_article slope_article
<fct>      <dbl>      <dbl>    <dbl>         <dbl>         <dbl>
1 pp        -0.259      0.898    0.124          0.422          0.672
2 ap         0.372      0.165    0.235          0.622          0.118
```

```
3 cp          0.366      0.123      0.447          0.475      0.097
# i 2 more variables: slope_ratio_ours <dbl>, slope_ratio_article <dbl>
```

Bibliography

- [1] A. Field, J. Miles, and Z. Field, *Discovering Statistics Using R*. London: SAGE Publications, 2012.
- [2] M. A. Lawrence, “ez: Easy Analysis and Visualization of Factorial Experiments.” 2026. doi: 10.32614/CRAN.package.ez.
- [3] S. Olejnik and J. Algina, “Generalized eta and omega squared statistics: Measures of effect size for some common research designs,” *Psychological Methods*, vol. 8, no. 4, pp. 434–447, 2003, doi: 10.1037/1082-989X.8.4.434.
- [4] R. Bakeman, “Recommended effect size statistics for repeated measures designs,” *Behavior Research Methods*, vol. 37, no. 3, pp. 379–384, 2005, doi: 10.3758/BF03192707.
- [5] H. Wickham *et al.*, “Welcome to the tidyverse,” *Journal of Open Source Software*, vol. 4, no. 43, p. 1686, 2019, doi: 10.21105/joss.01686.
- [6] D. Mottet, L. E. H. van Dokkum, J. Froger, A. Gouaich, and I. Laffont, “Trajectory formation principles are the same after mild or moderate stroke,” *PLoS ONE*, vol. 12, no. 3, p. e173674, 2017, doi: 10.1371/journal.pone.0173674.
- [7] P. M. Fitts, “The information capacity of the human motor system in controlling the amplitude of movement,” *Journal of Experimental Psychology*, vol. 47, no. 6, pp. 381–391, 1954, doi: 10.1037/h0055392.